

Hybrid θ -Continuation with r -Parameterization

1 Setup

We solve

$$\min_x c^T x + \frac{1}{2} x^T Q x \quad \text{s.t.} \quad Ax + b \succeq_K 0,$$

where K is a symmetric cone with identity e , quadratic representation $P(\cdot)$, and rank $m = \langle e, e \rangle$.

1.1 r -parameterization

Given a scaling point $W \succ 0$ and a centering vector $r \succ 0$, the primal slack and dual multiplier are

$$s = P(W^{-1/2})(r - \delta), \tag{1}$$

$$\lambda = P(W^{1/2})(r + \delta). \tag{2}$$

The complementarity gap is $\langle s, \lambda \rangle = \|r\|^2 - \|\delta\|^2$.

1.2 Homogenized self-dual embedding (HSE)

The variables $(x, \lambda, s, \tau, \theta)$ satisfy the HSE:

$$A^T \lambda = \tau c + Qx + \theta (A^T e - c), \tag{3}$$

$$s = Ax + \tau b + \theta (e - b), \tag{4}$$

$$b^T \lambda + c^T x + \frac{x^T Q x}{\tau} + \frac{\theta}{\tau} = \theta (\langle b, e \rangle + 1). \tag{5}$$

The parameter $\theta \in [0, 1]$ is a homotopy: at $\theta = 1$ the system has the trivial solution $(x, \lambda, s) = (0, e, e)$ with $\tau = 1$; at $\theta = 0$ it reduces to the original problem. The parameter τ controls homogenization.

2 Three-solve decomposition

2.1 KKT system and δ

At fixed (W, r, θ, τ) , we eliminate λ and s from the HSE (3)–(4) using the r -parameterization (1)–(2). Substituting $s = P(W^{-1/2})(r - \delta)$ into (4) gives

$$\delta = r - P(W^{1/2})(Ax + \tau b + \theta (e - b)),$$

and substituting $\lambda = P(W^{1/2})(r + \delta)$ into (3) yields, after eliminating δ , the normal equation

$$(A^T W^2 A + Q) x = 2 A^T P(W^{1/2})(r) - \tau (c + A^T P(W)(b)) - \theta (A^T e - c + A^T P(W)(e - b)). \quad (6)$$

Once x is obtained, δ is recovered from

$$\delta = r - P(W^{1/2})(A x + \tau b + \theta (e - b)). \quad (7)$$

2.2 Decomposing the RHS

The right-hand side of (6) separates into three independent terms:

$$\text{rhs}(\tau, \theta) = \underbrace{2 A^T P(W^{1/2})(r)}_{\text{rhs}_0} + \tau \underbrace{(-c - A^T P(W)(b))}_{\text{rhs}_1} + \theta \underbrace{(c - A^T e + A^T P(W)(b) - A^T P(W)(e))}_{\text{rhs}_2}.$$

Note that rhs_1 (coefficient of τ) and rhs_2 (coefficient of θ) use the original data (c, b) only. The blending of b and c arises from the combination $\tau \text{rhs}_1 + \theta \text{rhs}_2$, not from blended quantities b_θ or c_θ .

One can verify that $\text{rhs}_0 + \text{rhs}_1 + \text{rhs}_2$ vanishes at the fixed point ($W = I, r = e$):

$$\text{rhs}_0 + \text{rhs}_1 + \text{rhs}_2 = 2 A^T e - c - A^T b + c - A^T e + A^T b - A^T e = 0.$$

2.3 Three back-solves

Define three back-solves:

$$(A^T W^2 A + Q) x_0 = \text{rhs}_0 = 2 A^T P(W^{1/2})(r), \quad (8)$$

$$(A^T W^2 A + Q) x_1 = \text{rhs}_1 = -c - A^T P(W)(b), \quad (9)$$

$$(A^T W^2 A + Q) x_\theta = \text{rhs}_2 = c - A^T e + A^T P(W)(b) - A^T P(W)(e). \quad (10)$$

Then

$$x(\tau, \theta) = x_0 + \tau x_1 + \theta x_\theta,$$

and by (7),

$$\delta(\tau, \theta) = r - P(W^{1/2})(A x_0 + \tau (b + A x_1) + \theta (e - b + A x_\theta)).$$

Cancellation property. At the fixed point ($W = I, r = e$), $\text{rhs}_0 + \text{rhs}_1 + \text{rhs}_2 = 0$ (shown above), so

$$x_0 + x_1 + x_\theta = 0 \quad (\text{at } W = I, r = e). \quad (11)$$

At the fixed point ($\theta = 1, \tau = 1$) the lifted primal vanishes: $x(1, 1) = x_0 + x_1 + x_\theta = 0$.

3 Two-term decomposition

For fixed θ and r , define

$$f = x_0 + \theta x_\theta, \quad g = x_1. \quad (12)$$

3.1 Primal $x(\tau)$

From Section 2.3, $x(\tau, \theta) = x_0 + \tau x_1 + \theta x_\theta = (x_0 + \theta x_\theta) + \tau x_1$, so

$$\boxed{x(\tau) = f + \tau g.} \quad (13)$$

3.2 Direction $\delta(\tau)$

From (7), $\delta = r - P(W^{1/2})(Ax + \tau b + \theta(e - b))$. Substituting (13):

$$Ax + \tau b + \theta(e - b) = Af + \tau(Ag + b) + \theta(e - b).$$

Since $f = x_0 + \theta x_\theta$, the θ -dependent terms are absorbed into f . Defining

$$\delta_c = r - P(W^{1/2})(Af + \theta(e - b)), \quad (14)$$

$$\delta_t = -P(W^{1/2})(b + Ag), \quad (15)$$

we have

$$\boxed{\delta(\tau) = \delta_c + \tau \delta_t.} \quad (16)$$

3.3 Dual $\lambda(\tau)$

The homogenized dual is $\lambda = P(W^{1/2})(r + \delta)$. Defining

$$\lambda_c = P(W^{1/2})(r + \delta_c), \quad \lambda_t = P(W^{1/2})(\delta_t), \quad (17)$$

linearity of $P(W^{1/2})$ gives

$$\boxed{\lambda(\tau) = \lambda_c + \tau \lambda_t.} \quad (18)$$

3.4 Summary

Homogenized	
Primal	$x(\tau) = f + \tau g$
Direction	$\delta(\tau) = \delta_c + \tau \delta_t$
Dual	$\lambda(\tau) = \lambda_c + \tau \lambda_t$

where

$$\begin{aligned} f &= x_0 + \theta x_\theta, & g &= x_1, \\ \delta_c &= r - P(W^{1/2})(Af + \theta(e - b)), & \delta_t &= -P(W^{1/2})(b + Ag), \\ \lambda_c &= P(W^{1/2})(r + \delta_c), & \lambda_t &= P(W^{1/2})(\delta_t). \end{aligned}$$

3.5 Verification at the fixed point

At $(\theta = 1, \tau = 1, W = I, r = e)$: from (11), $x_0 + x_1 + x_\theta = 0$, so $f = x_0 + x_\theta = -x_1 = -g$ and $Af = -Ag$.

$$\begin{aligned}\delta_c &= e - (Af + (e - b)) = b - Af, \\ \delta_t &= -(b + Ag) = -(b - Af) = -\delta_c.\end{aligned}$$

Hence $\delta(1) = \delta_c + \delta_c \cdot 1 = \delta_c - \delta_c = 0$, $x(1) = f + g = 0$, and $\lambda(1) = P(W^{1/2})(e + 0) = e$. $\checkmark$

4 Tau as a cone variable

The homogenization parameter τ is treated as a scalar cone variable with its own scaling w_τ , centering parameter r_τ , and direction d_τ :

$$\tau = w_\tau r_\tau (1 + d_\tau). \quad (19)$$

The complementary variable is $\kappa = w_\tau^{-1} r_\tau (1 - d_\tau)$, giving $\tau \kappa = r_\tau^2 (1 - d_\tau^2)$.

Initialization: $w_\tau = 1, r_\tau = 1$.

5 Joint (τ, θ) selection

We select τ and θ simultaneously from two equations: the *gap equation* (duality identity) and the *normalization equation*.

5.1 The two equations

Both equations use the three-solve components $\lambda(\tau, \theta) = \lambda_0 + \tau \lambda_1 + \theta \lambda_\theta$ and $x(\tau, \theta) = x_0 + \tau x_1 + \theta x_\theta$ (before collapsing to the two-term form).

Gap equation. The duality identity in homogenized variables, with r_τ^2/τ replacing θ/τ :

$$b^T \lambda + c^T x + \frac{x^T Q x}{\tau} + \frac{r_\tau^2}{\tau} = \theta R, \quad (20)$$

where $R = \langle b, e \rangle + 1$. The substitution r_τ^2/τ decouples θ from the complementarity term; r_τ tracks the centering of the τ variable independently.

Normalization equation. At the fixed point $(\hat{x} = e, \hat{\lambda} = e, \hat{\tau} = 1, \hat{\theta} = 1)$, define

$$r_p = b - e, \quad r_d = c - A^T e, \quad r_g = -(b^T e + 1), \quad \alpha = \langle e, e \rangle + 1.$$

The normalization constraint is

$$r_p^T \lambda + r_d^T x + r_g \tau = -\alpha. \quad (21)$$

Substituting the three-component expansion:

$$N_0 + N_1 \tau + N_\theta \theta = -\alpha, \quad (22)$$

where

$$\begin{aligned} N_0 &= r_p^T \lambda_0 + r_d^T x_0, \\ N_1 &= r_p^T \lambda_1 + r_d^T x_1 + r_g, \\ N_\theta &= r_p^T \lambda_\theta + r_d^T x_\theta. \end{aligned}$$

Complementarity identity. At any point, the total complementarity satisfies

$$\underbrace{\|r\|^2 - \|\delta\|^2}_{\text{gap}} + r_\tau^2 = \theta \alpha. \quad (23)$$

5.2 Eliminating θ

From the normalization (22):

$$\theta(\tau) = \frac{-\alpha - N_0 - N_1 \tau}{N_\theta}. \quad (24)$$

Define $n_1 = N_1/N_\theta$ and $e_1 = (\alpha + N_0)/N_\theta$. Substituting $\theta(\tau)$ into (20) gives $x(\tau) = f_0 + \tau h$ where $f_0 = x_0 - e_1 x_\theta$, $h = x_1 - n_1 x_\theta$.

After substitution, the gap equation $\times \tau$ becomes the *violation quadratic*:

$$\boxed{\mathcal{A} \tau^2 + \mathcal{B} \tau + \mathcal{C} = 0}, \quad (25)$$

with

$$\mathcal{A} = b^T l_1 + c^T h + h^T Q h + R n_1, \quad (26)$$

$$\mathcal{B} = b^T l_0 + c^T f_0 + 2 f_0^T Q h + R e_1, \quad (27)$$

$$\mathcal{C} = f_0^T Q f_0 + r_\tau^2, \quad (28)$$

where $l_0 = \lambda_0 - e_1 \lambda_\theta$, $l_1 = \lambda_1 - n_1 \lambda_\theta$. The Q inner products ($f_0^T Q f_0$, $f_0^T Q h$, $h^T Q h$) are computed from the combined vectors f_0 and h , avoiding the six cross-terms of (x_0, x_1, x_θ) that suffer catastrophic cancellation when $x_1 \approx -x_\theta$.

5.3 Root selection and d_τ

Let τ' be a root of (25). Compute

$$d_\tau = \frac{\tau'}{w_\tau r_\tau} - 1. \quad (29)$$

Among the two roots, prefer the one with $\tau' > 0$; if both are positive (or both negative), pick the one with smaller $|d_\tau|$.

Recover θ from (24), then form $f = x_0 + \theta x_\theta$, $g = x_1$ and δ_c , δ_t as in Section 3.

Set $d_\infty = \max(\|d\|_\infty, |d_\tau|)$.

6 Algorithm: ThetaContinuationR

1. Initialize $W = I$, $r = e$, $\theta = 1$, $\tau = 1$, $w_\tau = 1$, $r_\tau = 1$.
2. Factor $(A^T W^2 A + Q)$. Compute the three-solve decomposition (8)–(10).
3. Solve the joint quadratic (25) for τ' . Compute d_τ from (29). Recover θ from (24). Form f , g , δ_c , δ_t .
4. Evaluate $\delta(\tau)$, solve Lyapunov for d , compute gap and $d_\infty = \max(\|d\|_\infty, |d_\tau|)$.
5. Check complementarity error: $\varepsilon_c = |\text{gap} + r_\tau^2 - \theta \alpha|$. If ε_c exceeds a threshold, freeze W (only r-updates allowed).
6. **W-update** (center): geodesic step on (W, r) with step size $\alpha = \min(1, 2/d_\infty^2)$, and $w_\tau \leftarrow w_\tau \exp(d_\tau \cdot \alpha)$. Refactor.
7. **r-update** (shrink): $r \leftarrow (r + |\delta|)/2$ (component-wise), and $r_\tau \leftarrow r_\tau (1 + |d_\tau|)/2$.
8. Repeat until $|\theta| < \varepsilon$, $|\text{gap}| < \varepsilon$, $d_\infty \leq 1$.

At convergence, recover $x_{\text{phys}} = x(\tau)/\tau$ and $\lambda_{\text{phys}} = P(W^{1/2})(r + \delta(\tau))/\tau$. The ratio $\kappa = \theta/\tau$ serves as an infeasibility certificate: $\kappa \rightarrow \infty$ indicates primal infeasibility ($\tau \rightarrow 0$).